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Department Of Software Engineering

Course: operating systems and systems programing

Group assignment

Group-6

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1. Introduction

In modern computing systems, hardware components and operating systems work together to deliver efficient performance, reliability, and scalability. Hardware provides the physical foundation of a computer system, while the operating system (OS) acts as an interface that manages hardware resources and enables software applications to function effectively.

This assignment focuses on analyzing four major hardware-related companies, each representing a critical component of computer systems. These include Huawei (HiSilicon), a leading designer of processors; MSI, a well-known manufacturer of computer systems; Nanya Technology, a key producer of memory (RAM); and TeamGroup, a company specializing in storage solutions. Each of these organizations contributes to a different layer of the computing architecture, forming a complete ecosystem.

Understanding the relationship between these hardware components and the operating system is essential for system design, performance optimization, and efficient programming. The processor executes instructions, memory stores active data, storage maintains long-term information, and the computer system integrates all components into a functional unit. The operating system coordinates these elements by managing resources, scheduling tasks, and providing an environment for applications.
The purpose of this study is to explore the architecture, evolution, and technological contributions of these companies, while also examining how their hardware interacts with operating systems and supports system-level programming.

2. Objectives

General Objective

The main objective of this assignment is to analyze the structure, functionality, and interaction of key computer hardware components—processor, computer system, memory, and storage—through the study of selected companies, and to understand their relationship with operating systems.

Specific Objectives

- To examine the architecture and design of processor systems, particularly Huawei HiSilicon processors
- To analyze the structure and components of computer systems developed by MSI
- To understand the role and functioning of memory systems, focusing on Nanya Technology

- To explore storage technologies and solutions provided by TeamGroup
- To study how operating systems interact with hardware components
- To evaluate hardware from a system programming perspective
- To analyze data flow within a computer system
- To compare technological advancements among the selected companies

3. Companies Under Study

3.1 Huawei (HiSilicon) – Processor

Huawei's HiSilicon is a semiconductor company responsible for designing advanced processor chips, particularly for smartphones and networking devices. Established as a subsidiary of Huawei, HiSilicon has played a major role in developing high-performance and energy-efficient processors such as the Kirin series. These processors are based on ARM architecture and integrate multiple components such as CPU, GPU, and AI modules into a single chip (SoC).

3.2 MSI (Micro-Star International) – Computer Systems

MSI is a global leader in the design and manufacture of computer hardware, especially gaming laptops, desktops, and motherboards. Known for high-performance systems, MSI integrates powerful CPUs, GPUs, and cooling technologies to create reliable and efficient computing environments.

3.3 Nanya Technology – Memory (RAM)

Nanya Technology is a major manufacturer of DRAM (Dynamic Random Access Memory), which is essential for temporary data storage in computer systems. The company focuses on high-speed and energy-efficient memory solutions used in computers, servers, and mobile devices.

3.4 TeamGroup – Storage

TeamGroup specializes in storage devices such as solid-state drives (SSDs), USB flash drives, and memory cards. Their products are designed to provide fast data access, reliability, and durability, which are critical for modern computing systems.

4. Processor System (Huawei HiSilicon)

4.1 Overview

The processor, also known as the Central Processing Unit (CPU), is the brain of the computer system. It executes instructions, performs calculations, and controls all other components. Modern processors are highly complex and include multiple cores, cache memory, and integrated processing units.

4.2 History and Growth

HiSilicon was established in 2004 as a semiconductor branch of Huawei. Over the years, it has evolved into a leading chip designer, particularly in mobile and networking processors. The introduction of the Kirin series marked a significant milestone, enabling Huawei to compete globally in smartphone technology.

4.3 Architecture and Design

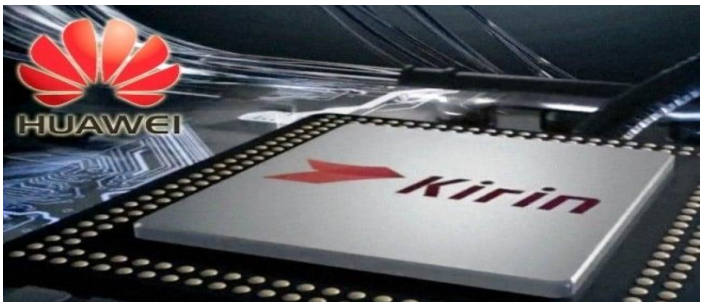
HiSilicon processors are primarily based on ARM architecture, which is known for power efficiency. These processors are designed as System-on-Chip (SoC), meaning they integrate:

- CPU cores (for general processing)
- GPU (for graphics processing)
- NPU (Neural Processing Unit for AI tasks)
- Cache memory

4.4 Products and Technologies

- Kirin series processors (e.g., Kirin 9000)
- AI-powered chips with machine learning capabilities
- 5G-enabled processors

These technologies allow faster processing, improved graphics, and enhanced connectivity.



4.5 Operating System Interaction

The operating system interacts with the processor through:

- Process scheduling
- Interrupt handling
- Resource allocation

The CPU executes OS instructions and manages multiple tasks efficiently using scheduling algorithms.

4.6 System Programming Perspective

From a programming perspective, the processor:

- Executes machine-level instructions
- Supports multithreading and multiprocessing
- Handles system calls from the OS

Understanding processor architecture helps developers write efficient and optimized programs.

5. Computer System (MSI – Micro-Star International)

5.1 Overview of Computer Systems

A computer system is an integrated combination of hardware and software components that work together to perform computational tasks. It includes the processor (CPU), memory (RAM), storage devices, input/output devices, and system software such as the operating system.
MSI (Micro-Star International) is recognized for designing high-performance computer systems, particularly in gaming, professional workstations, and enterprise environments. Their systems are engineered to deliver speed, stability, and efficient thermal performance, making them suitable for intensive computing tasks.

5.2 History and Growth of MSI

MSI was founded in 1986 in Taiwan and has grown into a globally recognized brand in computer hardware manufacturing. Initially focused on motherboards and graphics cards, the company expanded into complete computer systems such as laptops, desktops, and all-in-one PCs.
Over the years, MSI has built a strong reputation in the gaming industry by producing high-performance systems with advanced cooling solutions and powerful hardware configurations. Its continuous innovation in design and performance optimization has positioned it as a leader in competitive computing markets.

5.3 Hardware Components of MSI Computer Systems

- MSI computer systems are composed of several key hardware components:
- Central Processing Unit (CPU): Executes instructions and manages system operations
- Graphics Processing Unit (GPU): Handles graphics rendering and parallel processing
- Motherboard: Connects all components and enables communication between them
- Random Access Memory (RAM): Stores temporary data for quick access
- Storage Devices: Includes SSDs and HDDs for long-term data storage
- Cooling Systems: Advanced thermal solutions such as heat pipes and fans to maintain performance
- Power Supply Unit (PSU): Provides electrical power to all components

These components are carefully integrated to ensure optimal performance and reliability.

5.4 Products and Technologies

MSI offers a wide range of products, including:

- Gaming laptops and desktops
- High-performance motherboards
- Graphics cards (GPUs)
- Workstations for professional use

Key technologies include:

- Advanced cooling systems (e.g., Cooler Boost technology)

- Overclocking capabilities for enhanced performance
- High-refresh-rate displays in laptops
- Optimized BIOS and firmware systems

These innovations improve system speed, efficiency, and user experience.

5.5 Operating System Interaction

The operating system plays a crucial role in managing MSI computer systems by acting as an interface between hardware and software.

Key interactions include:

- Device Drivers: Enable communication between the OS and hardware components such as GPU, keyboard, and storage
- Process Management: The OS allocates CPU time to different applications
- Memory Management: Controls how RAM is used and shared among programs
- File System Management: Organizes and accesses data stored in storage devices
- Hardware Abstraction: Allows applications to run without directly interacting with hardware

Without the OS, MSI hardware would not function effectively as a complete system.

5.6 System Programming Perspective

From a system programming perspective, MSI computer systems provide a platform for low-level and high-level programming tasks.

Key aspects include:

- Direct hardware interaction: Programmers can interact with hardware through system calls and APIs
- Driver development: Writing software that controls hardware components
- Performance optimization: Efficient use of CPU, memory, and GPU resources
- Parallel processing: Utilizing multi-core CPUs and GPUs for faster execution

System programmers must understand the underlying hardware architecture to fully utilize the capabilities of MSI systems.

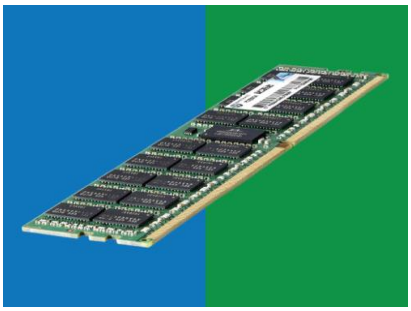
5.7 Role of MSI Systems in Modern Computing

MSI computer systems are widely used in:

- Gaming and entertainment
- Software development
- Artificial intelligence and machine learning
- Engineering and simulation

Their high-performance design makes them suitable for both personal and professional applications, demonstrating the importance of well-integrated computer systems in modern technology.

6. Memory System (Nanya Technology – RAM)



6.1 Overview of Memory Systems

The memory system is a critical component of a computer that stores data and instructions required for processing. It acts as a bridge between the processor and storage, providing fast access to actively used data.

Memory is generally classified into:

- Primary memory (Main Memory): RAM (volatile, fast access)
- Secondary memory: Storage devices like SSDs and HDDs (non-volatile, slower)

RAM (Random Access Memory) is especially important because it allows the CPU to quickly read and write data during program execution. Without sufficient memory, system performance significantly decreases.

6.2 History and Growth of Nanya Technology

Nanya Technology Corporation, established in 1995 in Taiwan, is one of the leading manufacturers of DRAM (Dynamic Random Access Memory). The company has continuously evolved its memory technologies to meet the growing demands of modern computing systems.

Nanya has focused on improving:

- Memory speed
- Power efficiency
- Storage capacity

Over time, it has contributed to advancements in computing devices such as personal computers, servers, and mobile systems.

6.3 Types of Memory and Technologies

Nanya primarily produces DRAM, but understanding memory types is essential:

- ➢ DRAM (Dynamic RAM):
- ✓ Stores data using capacitors
- ✓ Requires constant refreshing
- ✓ Used as main system memory
- ➢ SRAM (Static RAM):
- ✓ Faster and more expensive
- ✓ Used in CPU cache
- ➢ DDR SDRAM (Double Data Rate):
- ✓ Common modern RAM (DDR3, DDR4, DDR5)
- ✓ Transfers data on both clock edges for higher speed

Key Technologies by Nanya

- Advanced DRAM process technology
- High-density memory chips
- Low-power memory solutions

These technologies improve system performance and energy efficiency.

6.4 Memory Hierarchy

Memory systems are organized in a hierarchy based on speed, cost, and capacity:

1. Registers (fastest, smallest, inside CPU)
2. Cache Memory (L1, L2, L3)
3. Main Memory (RAM)
4. Secondary Storage (SSD/HDD)

This hierarchy ensures that frequently used data is stored closer to the CPU for faster access.

6.5 Operating System Interaction

The operating system plays a major role in managing memory efficiently.
Memory Allocation: Assigns memory space to processes
Key functions include:

- Virtual Memory: Uses disk space to extend RAM capacity
- Paging and Segmentation: Organizes memory into manageable units

- Memory Protection: Prevents processes from accessing each other's memory
- Caching: Improves speed by storing frequently accessed data

Efficient memory management ensures smooth multitasking and system stability.

6.6 System Programming Perspective

From a system programming point of view, memory management is crucial for performance and reliability.

Important concepts include:

- Pointers and memory addressing
- Dynamic memory allocation (e.g., malloc, free)
- Stack and heap memory usage
- Memory leaks and optimization

Programmers must use memory carefully to avoid inefficiencies and system crashes.

7. Storage System (TeamGroup)

7.1 Overview of Storage Systems

A storage system is responsible for retaining data permanently in a computer system. Unlike memory (RAM), which is volatile, storage devices preserve data even when the system is powered off.

Storage systems play a vital role in:

- Saving operating systems and applications
- Storing user data (documents, videos, etc.)
- Providing long-term data access

Modern storage systems are designed for high speed, reliability, and large capacity, making them essential for both personal and enterprise computing.

7.2 History and Growth of TeamGroup

TeamGroup Inc., founded in 1997 in Taiwan, is a well-known manufacturer of memory and storage products. The company has grown significantly by focusing on innovation, quality, and performance.

Initially producing basic storage devices, TeamGroup has expanded into advanced solutions such as:

- Solid-State Drives (SSDs)
- High-speed USB drives
- Memory cards

Today, it serves gamers, professionals, and general users with high-performance storage technologies.

7.3 Types of Storage Technologies

Storage systems can be categorized into different types:

1. Hard Disk Drive (HDD)
    - Uses magnetic disks to store data
    - Large capacity but slower speed
    - Mechanical parts make it less durable
2. Solid-State Drive (SSD)
    - Uses flash memory (no moving parts)
    - Much faster than HDD
    - More reliable and energy-efficient
3. Flash Storage
    - Used in USB drives and memory cards
    - Portable and convenient
    - Based on NAND flash technology

7.4 Products and Technologies by TeamGroup

TeamGroup provides a wide range of storage products:

- Internal SSDs (SATA & NVMe): High-speed storage for computers
- External SSDs: Portable and fast data transfer
- USB Flash Drives: Easy data sharing
- Memory Cards (SD, microSD): Used in mobile devices and cameras

Key Technologies

- 3D NAND Flash: Increases storage density and efficiency
- NVMe Technology: Provides faster data transfer through PCIe interface
- Wear Leveling: Extends lifespan of storage devices
- Error Correction Code (ECC): Ensures data reliability

7.5 File Systems and Storage Structure

Data in storage systems is organized using file systems, which are managed by the operating system.
Common file systems include:

- FAT32
- NTFS
- ext4 (Linux systems)

File systems:

- Organize data into files and directories
- Manage storage space
- Control access permissions

7.6 Operating System Interaction

The operating system interacts with storage systems through:

- File Management: Creating, reading, writing, and deleting files
- Disk Scheduling: Optimizing read/write operations
- Storage Allocation: Managing free and used space
- Device Drivers: Enabling communication with storage hardware

The OS ensures efficient and secure data handling across the system.

7.7 System Programming Perspective

From a system programming perspective, storage involves:

- File handling system calls (open, read, write, close)
- Buffering and caching techniques
- Disk I/O operations
- Data consistency and recovery mechanisms

Efficient storage programming improves performance and prevents data loss.

8. Data Flow in Computer System

8.1 Overview of Data Flow

Data flow in a computer system refers to the movement of data between different hardware components, including the processor, memory, storage, and input/output devices. This process is essential for executing programs and performing tasks efficiently.
A well-optimized data flow ensures Faster processing speed,Efficient resource utilization and Reduced system delays

8.2 Main Components Involved in Data Flow

Data flow occurs between the following key components:

- ✓ Processor (CPU): Executes instructions and controls operations
- ✓ Memory (RAM): Temporarily stores active data and instructions
- ✓ Storage (SSD/HDD): Holds permanent data
- ✓ Input Devices: Provide data to the system (keyboard, mouse)
- ✓ Output Devices: Display results (monitor, printer)

Each component plays a specific role in the movement and processing of data.

8.3 Step-by-Step Data Flow Process

To understand data flow clearly, consider the example of opening a file:

- User Input:

The user clicks a file using an input device.

- Os request

The operating system receives the request and identifies the file location.

- Storage Access:

The storage device (SSD/HDD) sends the requested data to RAM.

- Memory Loading:

Data is loaded into RAM for faster access.

- Processing:

The CPU fetches the data from RAM and processes it.

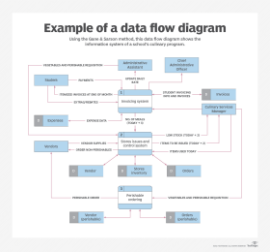
- Execution:

Instructions are executed (e.g., opening a document).

- Output:

The result is displayed on the screen.

This continuous movement of data ensures smooth system operation.



8.5 Interaction Between Components

Data flow is not linear; it is continuous and dynamic:

- ✓ CPU constantly reads/writes data from/to RAM

- ✓ RAM exchanges data with storage when needed
- ✓ Input/output devices communicate with the CPU via the OS

This interaction forms a cycle that keeps the system running efficiently.

8.6 Role of the Operating System in Data Flow

The operating system manages and controls data flow by:

- ✓ Scheduling processes for execution
- ✓ Managing memory allocation
- ✓ Controlling input/output operations
- ✓ Handling interrupts and signals

Without the OS, data flow would be uncoordinated and inefficient.

8.7 Factors Affecting Data Flow Performance

Several factors influence how efficiently data flows:

- ✓ CPU speed and number of cores
- ✓ RAM capacity and speed
- ✓ Storage type (SSD is faster than HDD)
- ✓ Bus width and speed
- ✓ System architecture

Optimizing these factors improves overall system performance.
Smooth output display

9. Integration of Hardware Components

9.1 Overview of Hardware Integration

Hardware integration refers to the coordination and interaction of all major computer components processor, memory, storage, and input/output devices—into a unified and functional system. Instead of working independently, these components operate together under the control of the operating system to execute tasks efficiently.
In a well-integrated system, each component complements the others, resulting in improved performance, reliability, and user experience.

9.2 Interaction Between Key Components

Processor and Memory

The processor (CPU) continuously interacts with RAM to fetch instructions and data required for execution. This interaction follows the fetch-decode-execute cycle, where:

- Instructions are fetched from memory
- Decoded by the CPU
- Executed to perform tasks

Faster memory leads to quicker data access, which improves CPU performance.

Processor and Storage

Storage devices such as SSDs provide long-term data, but they are slower compared to RAM. Therefore:

- Data is first transferred from storage to RAM
- The CPU then processes the data from RAM

This indirect interaction ensures efficiency and speed.

Memory and Storage

Memory and storage work together through techniques such as:

- Virtual Memory: Uses storage as an extension of RAM
- Caching: Stores frequently used data for faster access

This coordination helps in handling large programs and multitasking.

Input and Output Devices

Input devices send data to the system, while output devices present results to the user. The processor and OS manage this communication through:

- Device drivers
- Interrupt handling

9.3 Role of the Operating System in Integration

The operating system is the central controller that ensures all hardware components work together smoothly.

Its responsibilities include:

- Resource Management: Allocating CPU, memory, and storage
- Process Scheduling: Managing execution of tasks
- Device Management: Controlling hardware via drivers
- Communication Control: Coordinating data exchange between components

  Without the OS, hardware components would not function as a unified system.

10. Comparative Analysis

10.1 comparison of roles in computer system

company	Component type	Main role
Huawei(Hisilicon)	processor	Executes instructions and controls system operations

MSI	Computer system	Integrates all hardware in to complete system
Nanya technology	Memory (Ram)	Provides temporary high speed data storage
Teamgroup	storage	Provides long term data storage function

10.2 Technology Comparison

Huawei (HiSilicon)

- Focuses on ARM-based processor design
- Strong in AI integration and mobile processing
- High energy efficiency

MSI

- Focuses on high-performance computing systems
- Strong in gaming hardware and thermal design
- Uses advanced GPUs and CPUs in system builds

Nanya Technology

- Focuses on DRAM memory chips
- Emphasis on speed, efficiency, and density
- Competes in global semiconductor memory market

TeamGroup

- Focuses on SSD and flash storage technologies
- Uses NVMe and 3D NAND for high-speed performance
- Targets gaming and professional users

10.3 Performance Comparison

Processor (HiSilicon): Highest impact on computing speed and processing capability
Computer System (MSI): Depends on combination of all components for performance
Memory (Nanya): Affects multitasking and system responsiveness
Storage (TeamGroup): Impacts boot speed and data access time
Overall system performance depends on how balanced these components are.

11. Future Trends and Recommendations

11.1 Overview of Future Trends

The field of computer hardware is rapidly evolving due to increasing demand for faster processing, higher efficiency, and intelligent systems. Future developments in processors, memory, storage, and computer systems will be driven by artificial intelligence, miniaturization, energy efficiency, and high-speed connectivity.
The selected companies—Huawei (HiSilicon), MSI, Nanya Technology, and TeamGroup—are all expected to play a role in these advancements.

11.2 Future Trends in Processor Technology (Huawei HiSilicon)

Future processors are expected to become:

- More energy-efficient with lower power consumption
- Highly integrated with AI and machine learning capabilities
- Based on advanced architectures beyond current ARM designs
- Smaller in size but more powerful (advanced semiconductor nodes like 3nm and below)
- AI-driven chipsets for smart devices
- Improved 5G and 6G integration
- Enhanced System-on-Chip (SoC) designs

11.3 Future Trends in Computer Systems (MSI)

MSI computer systems are expected to evolve in the following ways:

- Greater focus on AI-powered performance optimization
- More compact but powerful gaming and workstation systems
- Advanced cooling technologies (liquid cooling and hybrid systems)
- Increased use of cloud-based computing integration

Gaming and professional systems will continue to require higher GPU and CPU performance with better energy efficiency.

11.4 Future Trends in Memory Systems (Nanya Technology)

Memory technology will continue improving in:

- DDR5 and DDR6 RAM development
- Higher memory bandwidth and speed
- Lower energy consumption
- Increased chip density (more storage in smaller physical space)
- High-performance DRAM for AI and big data systems
- More stable and energy-efficient memory chips
- Support for next-generation computing workloads

11.5 Future Trends in Storage Systems (TeamGroup)

Storage technology is moving toward:

- Faster NVMe SSDs with PCIe 5.0 and beyond
- Increased use of 3D NAND and possibly 4D NAND technologies
- Larger storage capacities in smaller physical devices
- Cloud-integrated storage solutions
- Ultra-fast SSDs for gaming and enterprise use
- Improved durability and lifespan of flash storage
- Better thermal management in high-speed drives

11.6 Final Insight: The future of computing will depend on how well hardware components evolve together. Faster processors, smarter memory systems, and high-speed storage will all work alongside advanced operating systems to create more intelligent and efficient computing environments

12. Conclusion

In conclusion, this study has provided a comprehensive analysis of four major hardware-related companies—Huawei (HiSilicon), MSI, Nanya Technology, and TeamGroup—and their roles in modern computer systems. Each company represents a critical component of computing architecture, including the processor, computer system, memory, and storage.

The processor system, represented by Huawei HiSilicon, is responsible for executing instructions and controlling overall system operations. MSI demonstrates how complete computer systems integrate multiple hardware components into a unified platform for high-performance computing. Nanya Technology highlights the importance of memory systems in enabling fast data access and efficient multitasking. Team Group illustrates the role of storage systems in ensuring long-term data retention and high-speed data retrieval.
Furthermore, the interaction between these components is coordinated by the operating system, which manages resources, schedules processes, and ensures smooth communication between hardware elements. The data flow within a computer system demonstrates how information moves efficiently from input to processing, memory, storage, and output.

Overall, this study shows that modern computing systems depend on the seamless integration of hardware components. Each component plays a unique role, but their combined functionality is what enables efficient, powerful, and reliable computing. Continuous advancements in processor technology, memory design, storage systems, and computer architecture will further enhance future computing performance

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